



Zoology Legends

CHE201

Final Term Papers Spring 2024

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[IgCpc6fBAQX](#)

Team Zoology LEGENDS



Zoology Legends

MCQS Spring 2024

(Quiz IN Pics/ Highlighted Must Prepare)

- ✓ Water is universal solvent due to **polarity**.
- ✓ Ideal gas equation is... **$Pv = nRT$**
- ✓ A key property in thermodynamic is **Temperature**.
- ✓ The quantitative study of heat changes is **Thermochemistry**.
- ✓ If we bring any change at equilibrium the system compensate for change this is called **Le Chateliers Principle**.
- ✓ The system in which heat change is negative is **endothermic**.
- ✓ Freezing point is a **cogitative** property.
- ✓ The amount of heat needed to increase the temperature of one gram of substance by degree is specific **heat capacity**.
- ✓ Two systems that are each found to be in thermal equilibrium with a third system will be found to be in thermal equilibrium with each other. This generalization from experience is **the Zeroth law of thermodynamics**
- ✓ Entropy of perfect crystalline substance is **zero at absolute zero**.
- ✓ A large number of experiments have determined that **4 variables** are sufficient to define the physical condition of a gas.
- ✓ When ΔG is negative, the forward reaction is **spontaneous**.
- ✓ Ideal gas obeys **Roult's law**.
- ✓ Oxidation of gold is example of **(non-spontaneous)**

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- ✓ Gibbs free energy is a.. (state function)
- ✓ **Mechanical Equilibrium**: No unbalanced forces act on or within the system; hence the system undergoes no acceleration, and there is no turbulence within the system.
- ✓ The macroscopic part of the universe under study in thermodynamics is called the System__**Thermodynamic**
- ✓ Charles' law is obeyed most accurately in the limit of zero **pressure**
- ✓ van der Waals in (1873) modified the ideal-gas equation
- ✓ The change in enthalpy is then equal to the heat transferred at constant (**pressure**)
- ✓ Entropy of a perfect crystalline substance is (**zero**) at absolute zero
- ✓ Throttling is example of (**irreversible process**)
- ✓ A large number of experiment have determined that__**4**__variables are sufficient to define the physical condition of a gas?
- ✓ Non-zero volumes of gas particles (**repulsive effect**)
- ✓ Attractive forces between gas particles (**attractive effect**)
- ✓ The amount of heat required to increase the temperature of one mole of a substance by one degree__**Molar heat capacity**
- ✓ if $k > 1$ it shows..... **Products** ...
- ✓ Quantative study of heat change is called thermochemistry
- ✓ aik water ki koi value pochi thi ans *41*tha

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- ✓ Principle which states that any change to a system at equilibrium will adjust to compensate for that change___**Le Châtelier's**
- ✓ Manfred Eigen (Germany) pioneered when, in the early 1960's, he measured the rate constant of what was then the---- reaction ever observed.___**Fastest**
- ✓ water is universal solvent due to ? **polarity**
- ✓ oxidation of gold is ? **non spontaneous process**
- ✓ carbonate and phosphate are **insoluble**
- ✓ The interactions between the solute particles and the solvent molecules is called___**Solvation**
- ✓ Shine a light beam through colloids the pathway of the light is visible from any angle. This scattering of light is called the ___**Tyndall effect**
- ✓ The amount of moles of solute per kilogram of the solvent___**Molarity**
- ✓ Activity is more accurate in more ---- solutions___**concentrated**
- ✓ Determines the real chemical potential for a real solution rather than an ideal___**Activity**

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- ✓ If ΔG is positive then the reaction will be....(non_spontaneous)
- ✓ If $K < 1$ then it favors (reactants)
- ✓ Thermodynamics studies the relationships between the ____ of a system.... (Macroscopic properties)
- ✓ The interactions between the solute particles and the solvent molecules is called _____. (Solvation)
- ✓ Continuous Flow is a type of assay used for determining initial value and _____. (Inhibition values)
- ✓ A first-order reaction is a reaction that proceeds at a rate that depends on _____ (One reactant) concentration
- ✓ Boiling point of Ethanol is _____. (78.5°C)
- ✓ The number of particles involved in an elementary step is called _____. (Molecularity)
- ✓ The fan in a 'frost free' home freezer moves air around inside the freezer to _____. (Sublimate)
- ✓ (Fog and mist) are examples of colloidal liquid in a gas.
- ✓ Most hydroxide salts are only slightly soluble, except those of... (sodium and potassium)
- ✓ ethanol will start to boil at 78.4 °C

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- ✓ zero order reactions/kinetics, the rate of a reaction does not depend on the ...(substrate concentration)
- ✓ The activation energy is the threshold energy that the reactant(s) ...(must acquire before reaching the transition state)
- ✓ Once started, chain reactions continue until the.... (reactants are exhausted)
- ✓ mass/volume percentage are...(relative concentration unit)
- ✓ The epitome of intermolecular forces in solution is the miracle of solubility, because when a matter precipitates it no longer interacts with the(solvent.)
- ✓ The macroscopic part of the universe under study in thermodynamics is called the (System)
- ✓ from the stoichiometry of the reaction equation the order should not determined
- ✓ The number of particles involved in an elementary step is called_Molarity
- ✓ In zero order reactions/kinetics, the rate of a reaction does not depend on the substrate concentration
- ✓ Freezing point is _cogitative_ property

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- ✓ The amount of moles of solute per liter of the solution is **Molarity**
- ✓ The formation of double stranded DNA from two complementary strands, can be described using **2nd order kinetics**
- ✓ continues flow is commonly used for determining initial rates and **inhibition values**
- ✓ The rate at which the reagents are first brought together is **initial rate of reaction**
- ✓ Concentrated form of ethanol, an azeotrope, is around **95.6 ethanol** by weight
- ✓ Boiling point of pure ethanol is **78.5**
- ✓ Measurement of relative amount of products and reactant present at a given time is called? **Reaction kinetics**
- ✓ Any change to a system at equilibrium will adjust to compensate for that change is? **Le Chatelier's principle**
- ✓ Water is a good solvent due to? **Its ability to dissolve a wide range of substances**
- ✓ The interaction between solute particle and solvent molecule is called? **Solvation**

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- ✓ In zero order reaction the rate of reaction does not depend? **On the concentration of the reactants**
- ✓ Freezing point is aproperty ? **heat**
- ✓ A key property in thermodynamics is.....? **temperature**
- ✓ Two thermodynamics system in equilibrium have same? **Temperature**
- ✓ Thermodynamics studies the relationships between the ___of a system. **Macroscopic properties**
- ✓ For a system in mechanical equilibrium, the pressure throughout the system is. **Uniform**
- ✓ A measurement of the relative amounts of products and reactants present in a reaction at a given time is called. **Reaction quotient**
- ✓ The interactions between the solute particles and the solvent molecules is called_. **Solvation**
- ✓ Continuous Flow is a type of assay used for determining initial value and_. **Inhibition values**
- ✓ The number of particles involved in an elementary step is called_. **Molecularity**
- ✓ The fan in a 'frost free' home freezer moves air around inside the freezer to_. **Sublimate**

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- ✓ A first-order reaction is a reaction that proceeds at a rate that depends on. **One reactant concentration**
- ✓ The rate at which the reagents are first brought together is called ___. **initial rate**
- ✓ The relaxation method is used to determine the rate of reaction of ___. **return to equilibrium.**
- ✓ A spectrometer is a device used to ___. **the amount of photons (the intensity of light)**
- ✓ A first order reaction is a reaction that depends on ___. **linearly on only one reactant concentration.**
- ✓ Once started a chain reaction continues until the ___. **reactants are exhausted**
- ✓ The rate constant is calculated by integrated rate laws from initial concentration and ___. **time of measurement.**
- ✓ Length of half life will be constant, **independent of concentration.**
- ✓ Another way to see it is that the half life of a first order reaction is independent of its **initial concentration.**
- ✓ For 1st and 2nd order reaction the graph will be ___ **decrease to zero with time**

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- ✓ The slow step is mechanism determines the __. **rate of a reaction**
- ✓ The number of particles involved in an elementary steps is called __. **molecularity**
- ✓ if ΔG is negative the reaction will be __. **spontaneous**
- ✓ The freezing point of solution is directly proportional to the __ of the solute. **molality**
- ✓ In __ van der Waals modified the ideal Gas. **1873**
- ✓ __ are liquid or solid with a Gas dispersed into them.

Foams

- ✓ Continuous Flow is a type of assay used for determining initial value and __. **Inhibition values**
- ✓ A measurement of the relative amounts of products and
- ✓ reactants present in a reaction at a given time is called __ **Reaction quotient**
- ✓ Thermodynamics studies the relationships between the
- ✓ __ of a system. **Macroscopic properties**
- ✓ Concentrated form of ethanol, an azeotrope, is around **95.6** ethanol by weight.
- ✓ continuous flow is commonly used for determining initial
- ✓ rates and __ **inhibition values**.

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- ✓ The fan in a 'frost free' home freezer moves air around inside the freezer to_. **Sublimate**
- ✓ Half life of 1st order -----**0.693**
- ✓ The fractionating column is packed with glass beads to give maximum **surface area** for vapor to condense on.
- ✓ Freezing points is a___ property. **Colligative**
- ✓ Per liter walay ka **Molarity**
- ✓ Per kilogram walay ka **Molality**
- ✓ Ideal gas equation is... **$Pv = nRT$**
- ✓ key property in thermodynamic is (**Temperature**)
- ✓ Gibbs free energy is a.. (**state function**)
- ✓ van der Waals in (**1873**)
- ✓ The interactions between the solute particles and the solvent molecules is called (**Solvation**)
- ✓ Thermodynamics studies the relationships between the of a system..(**Macroscopic properties**)
- ✓ The number of particles involved in an elementary step is
- ✓ called (**Molecularity**)
- ✓ The fan in a 'frost free' home freezer moves air around inside the freezer to (**Sublimate**)

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- ✓ Most hydroxide salts are only slightly soluble, except those of (**sodium and potassium**)
- ✓ activation energy is the threshold energy that the reactant. (**must acquire before reaching the transition state**)
- ✓ Once started, chain reactions continue until the (**reactants are exhausted**)
- ✓ The vapor density of C_xH_x is 39. The value of X is....? **3**
- ✓ Highest hydrogen ion concentration....? **lowest pH value.**
- ✓ if $\Delta H_{\text{solution}} = 0$ then the solution are called.....? **ideal solution**
- ✓ in.....van der wals modified the ideal Gass. **1873**
- ✓ **Heat mixture** containing 2 or more substances (e.g., water and ethanol solution) **fractional Distillation**
- ✓ Ethanol will start to boil at **78.4 °C (351.6 K)** and water at **100 °C (212 °F)**.
- ✓ If $K > 1$ then equilibrium favors **products**
- ✓ If $K < 1$ then equilibrium favors the **reactants**
- ✓ Most chemical chain reactions have very reactive intermediates called **free radicals**.

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- ✓ The intermediate that maintains the chain reaction is called a **chain carrier**.
- ✓ All nitrates (NO_3^-), most sulfates, (SO_4^{2-}), and most chlorides, Cl^- , are **soluble**.
- ✓ Most hydroxides, (OH^-), carbonates, (CO_3^{2-}) sulfides, S^{2-} , and phosphates, (PO_4^{3-}), are **insoluble**.
- ✓ An example of ideal solutions would be **benzene and toluene**.
- ✓ A maximum-boiling point azeotrope is said to be a **negative azeotrope**.
- ✓ The interaction between solute particles and solvent -----
----- **Solvation**
- ✓ The **fractionating** column is packed with glass beads (or something similar) to give the maximum **possible surface area** for vapor to condense on.
- ✓ Any change to a system at equilibrium will adjust to compensate for that change is? **Le Châtelier's principle**
- ✓ Water is a good solvent due to? **Polarity**
- ✓ In zero order reaction the rate of reaction does not depend? **Substrate concentration**.

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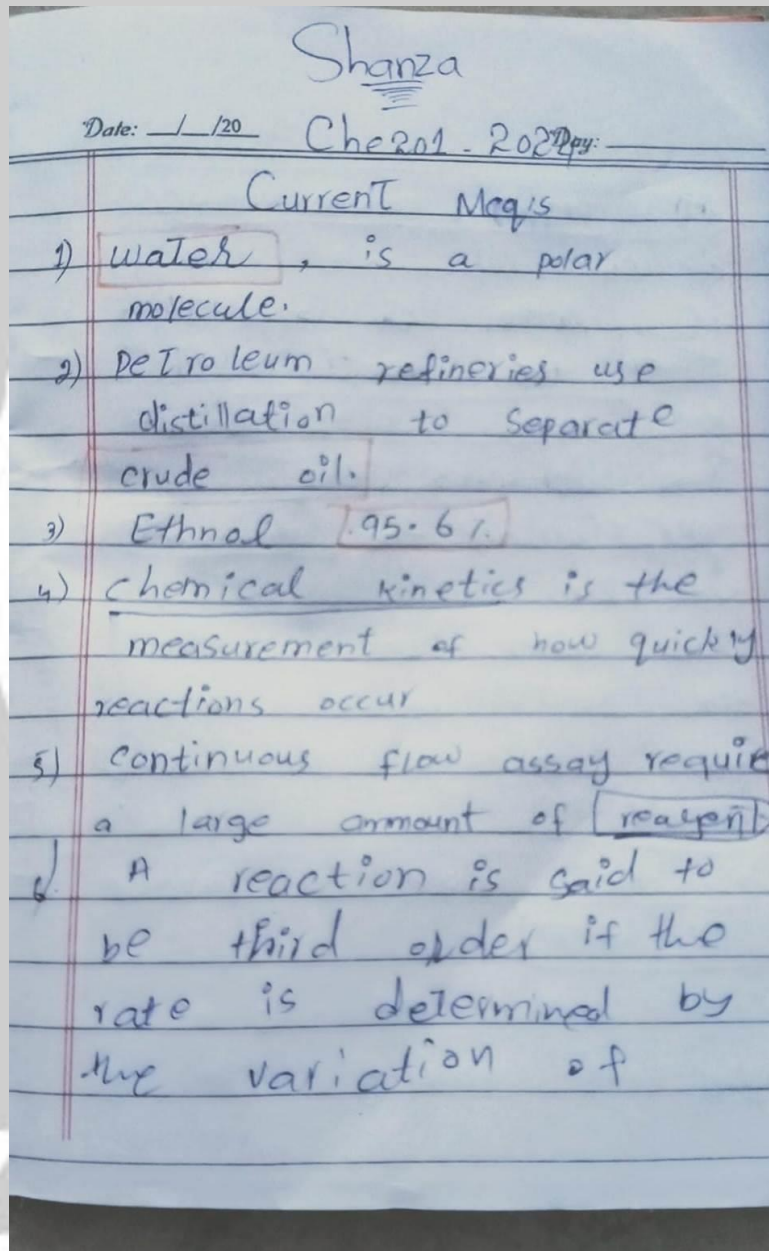
- ✓ Total amount of energy of non-spontaneous reaction is... is a **loss and negative** net result
- ✓ When **volume** is increased the change occurs in the direction that produces more moles of **gas**.
- ✓ The amount of moles of solute per liter of the solution__
Molarity
- ✓ Solute separates into ions or molecules, and each ion or molecule is surrounded by molecules of solvent...
dissolution
- ✓ Most hydroxide salts are only slightly soluble, except those of **sodium and potassium**.
- ✓ Most sulfide, carbonate and phosphate salts are only **slightly soluble**.
- ✓ Spectrophotometry is one of the most useful methods that use to measure how much a chemical substance absorbs light by **measuring the intensity of light**

(Note: Neachay or B hain MCQS Must Do IT)

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Date: / / 20

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of three concentration terms.

7) if ΔG is positive then the reaction will be (non spontaneous)

8) Thermodynamics the relationship the (Microscopic) properties

9) The interactions between the solute particles and solvent molecules is called Solvation.

10) if $K < 1$ then it favours (Reactants)

11) continuous flow is a type of assay used



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for determining initial value and — (inhibition values)

12) A first order reaction is a reaction that proceeds at a rate that depends on (one reactant)

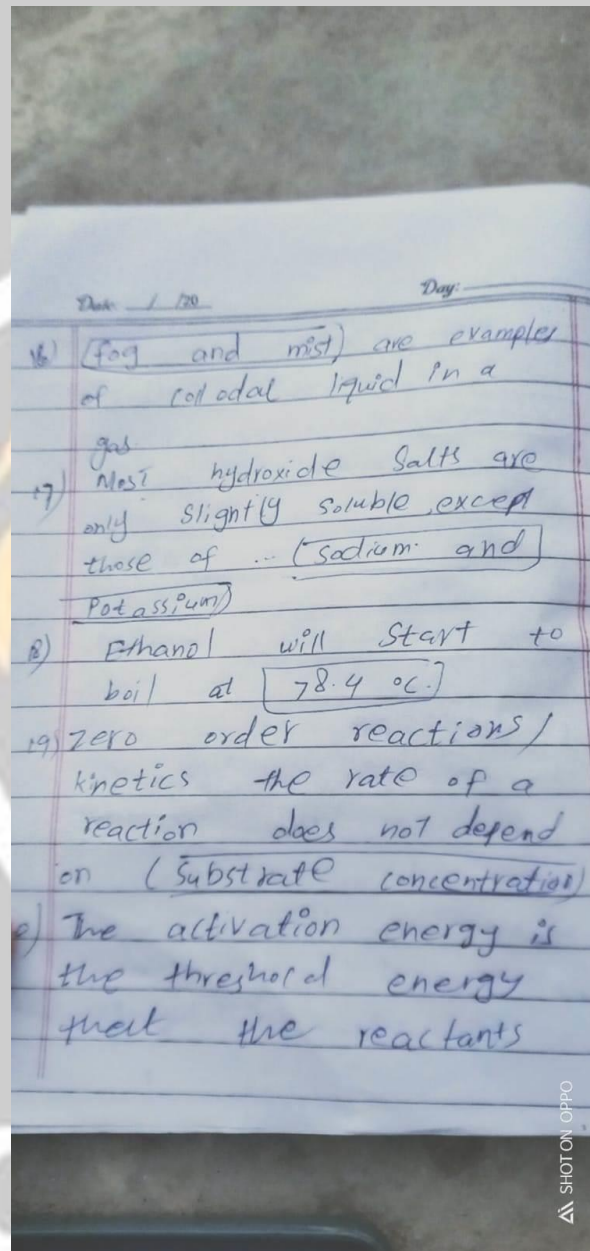
13) Boiling Point of Ethanol is (78.5 °C)

14) Number of particles involved in a elementary step is called (Molecular)

15) The fan in a frost free home freezer moves air around inside the freezer to — (Sublimation)



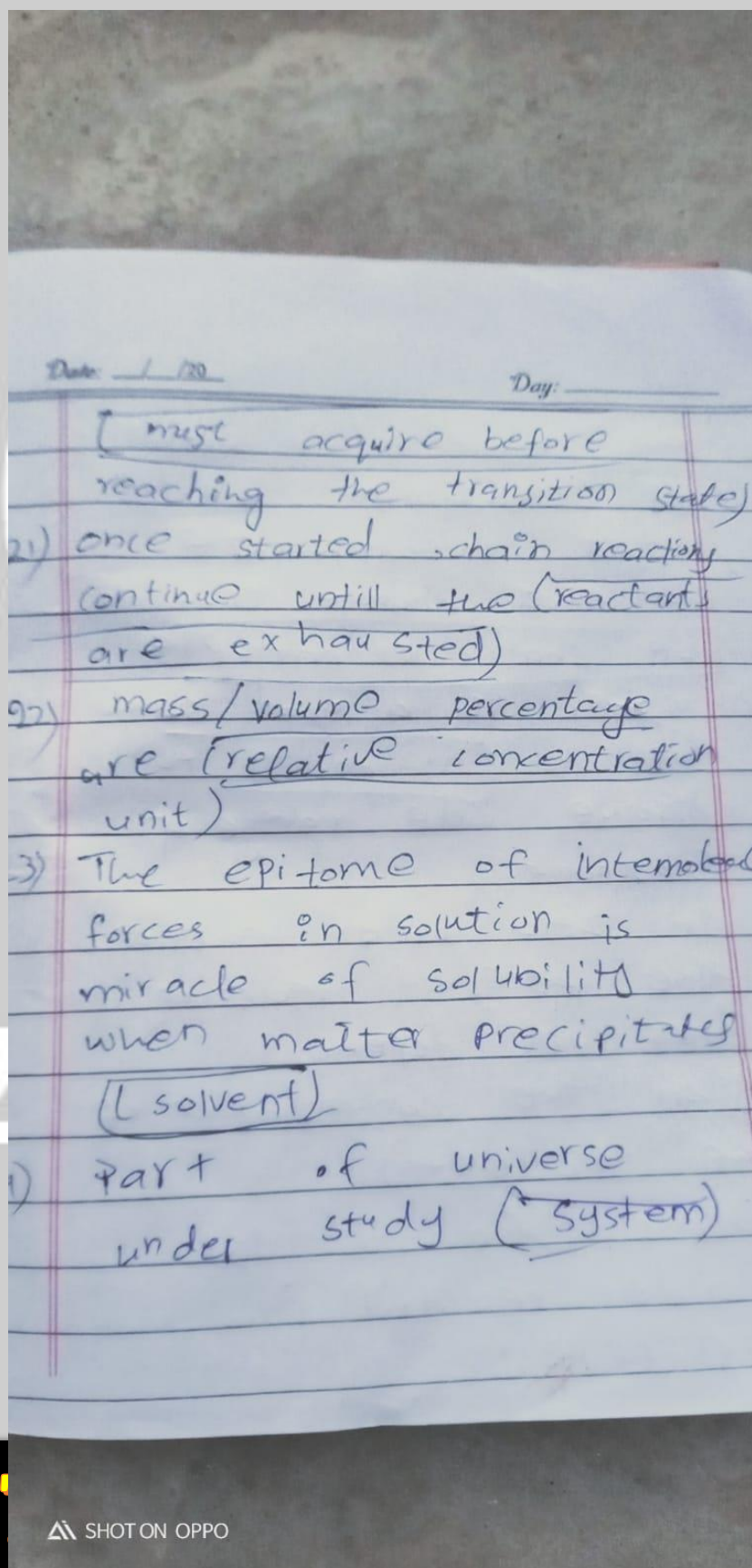
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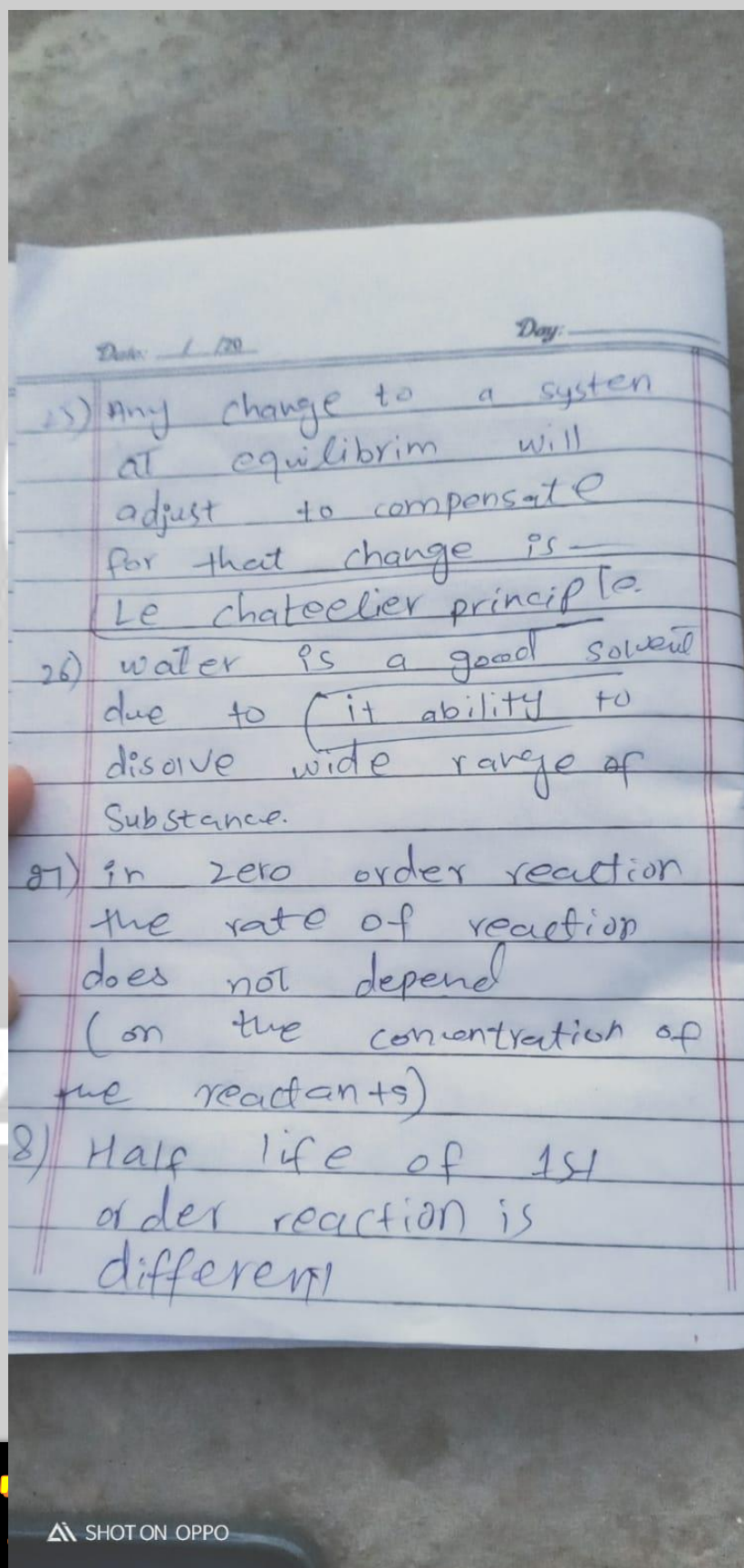


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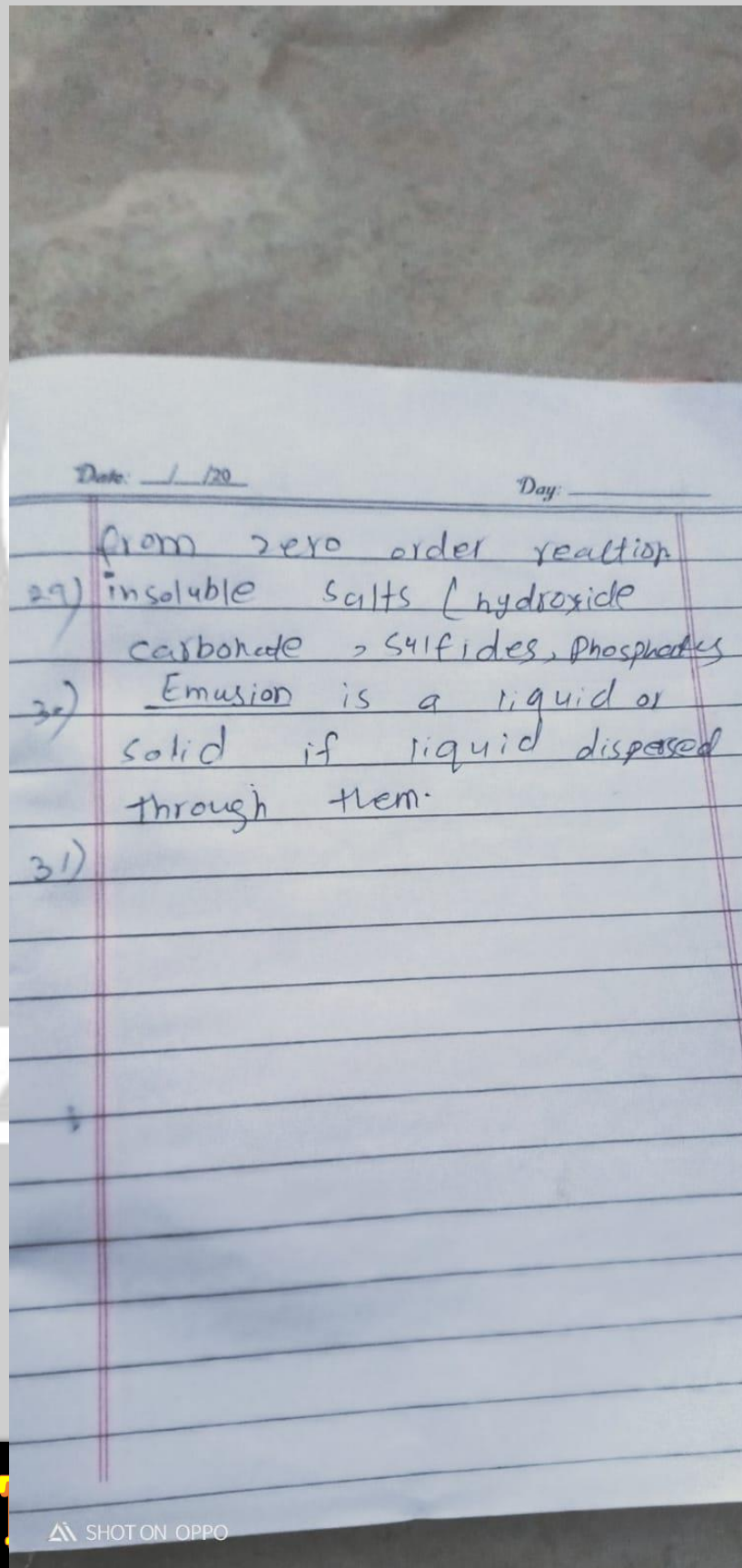


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1. Thermodynamics studies the relationships between the _____ of a system. **Macroscopic properties**
2. Two systems in thermal equilibrium with each other have the same _____. **Temperature**
3. For a system in mechanical equilibrium, the pressure throughout the system is _____. **Uniform**
4. According to Charles law the volume of a fixed amount of gas, at constant pressure, is proportional to the _____. **Absolute temperature**
5. Calorimetry is _____. **Measurement of Enthalpy**
6. A measurement of the relative amounts of products and reactants present in a reaction at a given time is called _____. **Reaction quotient**
7. Number of moles of solute per liter of the solution _____. **Molarity**
8. _____ are liquids or solids with a gas dispersed into them. **Foams**
9. The interactions between the solute particles and the solvent molecules is called _____. **Solvation**
10. _____ is often used as a way to purify a compound. **Precipitation**
11. In fractional distillation fractionating column is packed with glass beads to give _____. **Maximum possible surface area for vapor to condense on**
12. Continuous flow is a type of assay used for determining initial value and _____. **Inhibition values**
13. The rate at which the reagents are first brought together is called _____. **Initial rate**
14. The relaxation methods can be used to determine the _____. **Rate of the reaction**
15. A first-order reaction is a reaction that proceeds at a rate that depends on _____. **One reactant concentration**
16. Boiling point of Ethanol is _____. **78.5°C**

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17. The number of particles involved in an elementary step is called _____. **Molecularity**
18. The fan in a 'frost free' home freezer moves air around inside the freezer to _____. **Sublimate**
19. Extensive properties depend on the _____ of the system. **Parts**
20. A non-spontaneous is also known as _____. **Endergonic reaction**
21. _____ influences the physical properties of liquids. **Intermolecular forces**



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Water, a polar molecule



Third Order Reaction

A reaction is said to be of third order if the rate is determined by the variation of three concentration terms. In other words, the

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← CHE201 Final Terms Current Fall ...

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M.C.Q

1. Thermodynamics studies the relationships between the _____ of a system. **Macroscopic properties**
2. Two systems in thermal equilibrium with each other have the same _____. **Temperature**
3. For a system in mechanical equilibrium, the pressure throughout the system is _____. **Uniform**
4. According to Charles law the volume of a fixed amount of gas, at constant pressure, is proportional to the _____. **Absolute temperature**
5. Calorimetry is _____. **Measurement of Enthalpy**
6. A measurement of the relative amounts of products and reactants present in a reaction at a given time is called _____. **Reaction quotient**
7. Number of moles of solute per liter of the solution _____. **Molarity**
8. _____ are liquids or solids with a gas dispersed into them. **Foams**
9. The interactions between the solute particles and the solvent molecules is called _____. **Solvation**
10. _____ is often used as a way to purify a compound. **Precipitation**
11. In fractional distillation, each rectifying column is packed with glass beads to give _____. **Maximum possible surface area for vapor to condense on**
12. Continuous assay is a type of assay used for determining initial value and _____. **Inhibition values**
13. The rate at which the reagents are first brought together is called _____. **Initial rate**
14. Relaxation methods can be used to determine the _____. **Rate of the reaction**
15. A first-order reaction is a reaction that proceeds at a rate that depends on _____. **One reactant concentration**
16. Boiling point of Ethanol is _____. **78.5°C**

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17. The number of particles involved in an elementary step is called _____. **Molecularity**
18. The fan in a 'frost free' home freezer moves air around inside the freezer to _____. **Sublimate**
19. Extensive properties depend on the _____ of the system. **Parts**
20. A non-spontaneous is also known as _____. **Endergonic reaction**
21. _____ influences the physical properties of liquids. **Intermolecular forces**

Subjective

Qno1: What is the effect of catalyst on the activation energy? **2 marks**

If a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is formed.

Qno2: What is pseudo first order reaction? **2 marks**



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- d) none
- 180) The number of particles involved in an elementary step is called
- a) **Molecularity**
 - b) Order of reaction
 - c) Concentration of reactants
 - d) None
- 181) The equation in an elementary step represents the reaction
- a) **at the molecular level**
 - b) overall reaction
 - c) both



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7. K_c is written

- a) With units
- b) **Without units**
- c) None

8. If $K < 1$ then it favors

- a) **Reactants**
- b) Products
- c) Will be at equilibrium
- d) None

9. If $Q > K_c$ then which reaction will occur

- a) Forward
- b) **Reverse**

10. If $Q < K_c$ then which reaction will occur

- a) **Forward**
- b) Reverse

11. If ΔG is negative then the reaction will be

- a) Non spontaneous
- b) **Spontaneous**
- c) Both

12. If ΔG is negative then

- a) K would be very small
- b) **K would be very large**
- c) Both

13. If ΔG is positive then the reaction will be

- a) **Non spontaneous**
- b) Spontaneous
- c) Both

14. At standard conditions activities of all reactants and products is not equal to 1

- a) **True**
- b) False

15. If ΔG is positive then the----- reaction will be spontaneous.

- a) **Reverse**
- b) Forward
- c) Equilibrium

WU Medical Zone effort by Amara

1

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Zoology Legends

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he201 Current OBjective+Subjective papers for final term fall 20

- 1) Water is universal solvent due to-----? **polarity**
- 2) Oxidation of gold is -----? **Non spontaneous process**
- 3) Two systems that are each found to be in thermal equilibrium with a third system will be found to be in thermal equilibrium with each other is law? **Zeroth law**
- 4) In calorimetry we study the change in? **heat**
- 5) Freezing point is aProperty? **cogitative**
- 6) The magnitude of force that controls the shape of the liquid is called? **Surface tension**
- 7) The dipole moment is found in? **I guess NaCl or Water**
- 8) The amount needed to increase the temperature of one mole of a substance is called? **Molar heat capacity**
- 9) Gibbs free energy is a? **state function**
- 10) Fugacity Measuresof a Gas. **Non ideality**
- 11) $Pv = nRT$? **Ideal gas equitation.**
- 12) A key property in thermodynamics is-----? **Temperature**
- 13) Two thermodynamics systems in equilibrium have same -----? **Temperature**
- 14) If we bring any change at equilibrium the system compensate for change this is called? **Le Chatelier's Principle**
- 15) The mass/mass percentage is among the-----? **Relative amounts**
- 16) The quantitative study of heat changes is called-----? **Thermochemistry**
- 17) Density and pressure are examples of is -----? **Intensive thermodynamic property**
- 18) The system in which heat change is positive is? **Exothermic**

19) Define physical chemistry?
Physical Chemistry:
Physical chemistry is the study of the underlying physical principles that govern the properties and behavior of chemical systems.

20) What is the effect of the catalyst on the of activation energy?
However, if a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is formed. Enzymes can be thought of as biological catalysts that lower activation energy.

- A catalyst provides a reaction route with a lower activation energy.
- Suppose that the catalyzed activation energy is to 25 kJ mol⁻¹.
- The calculation is represented at 293 K:

$$e^{\frac{E_A}{RT}} = e^{\frac{25000}{8.31 \times 293}}$$
$$= 3.47 \times 10^{-5}$$

- Compared with the corresponding value for activation energy of 50 kJ mol⁻¹, there is significant increase in the fraction of molecules able to react.

reated By Hafiza Mubeen And Muhammad Nouman'efforts. **Page**

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12:04 PM

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Question # 10 of 10 (Start time: 11:06:11 PM, 20 January 2024) Total Marks: 1

Most sulfide, carbonate and phosphate salts are -----?

Select the correct option

- ☐ Sparingly soluble
- ☒ Insoluble
- ☐ Soluble
- ☐ Slightly soluble

[Show Answer](#)

[Click to Save Answer & Move to Next Question](#)

10:07 AM 1/20/2024

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12:03 PM

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Question # 8 of 10 (Start time: 09:24:44 PM, 20 January 2024) Total Marks: 1

By making changes in the reaction conditions and measuring the effect of the changes on the rate of reaction, we can infer what is going on at the_____.

Select the correct option

- ☒ molecular level
- ☐ electronic level
- ☐ subatomic level
- ☐ atomic level

Show Answer

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Water, a polar molecule



Petroleum refineries use distillation to separate crude oil

- For gases, experimenters use a buret to measure the change in volume produced at different times. They then relate these volumes to changes in concentration. •

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11:59 AM

20

CHE 201 Current Final terms Fall 2020

M.C.Q

1. Thermodynamics studies the relationships between the _____ of a system. **Macroscopic properties**
2. Two systems in thermal equilibrium with each other have the same _____. **Temperature**
3. For a system in mechanical equilibrium, the pressure throughout the system is _____. **Uniform**
4. According to Charles law the volume of a fixed amount of gas, at constant pressure, is proportional to the _____. **Absolute temperature**
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10. _____ is often used as a way to purify a compound. **Precipitation**
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16. Boiling point of Ethanol is _____. **78.5°C**

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17. The number of particles involved in an elementary step is called _____. **Molecularity**
18. The fan in a 'frost free' home freezer moves air around inside the freezer to _____. **Sublimate**
19. Extensive properties depend on the _____ of the system. **Parts**
20. A non-spontaneous is also known as _____. **Endergonic reaction**
21. _____ influences the physical properties of liquids. **Intermolecular forces**

Subjective

Qno1: What is the effect of catalyst on the activation energy? 2 marks

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LET FORT BY AMAAN

VU Med

CHE201:Grand Quiz

Question # 27 of 30 (Start time: 12:55:53 AM, 26 December 2020)

The quantitative study and measurement of heat and enthalpy changes is known as _____.

Select the correct option

☒	Thermochemistry
☐	Physical Chemistry
☐	Thermodynamics
☐	Organic Chemistry

Answer Solved by Amaan KI
Contact: 0305-4716616
Email: maniamaan2@gmail.

Windows icons: Internet Explorer, File Explorer, VLC, Chrome

Annotation tools: Text, Erase, Lasso, Highlight, Stamp, Checkmark, X

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11:55 AM

CHE201:Grand Quiz

Question # 4 of 30 (Start time: 01:19:26 AM, 26 December 2020)

If ΔG is a negative number the said reaction is

Select the correct option

☐	Non spontaneous
☐	Endothermic
☐	Exothermic
☒	Spontaneous

Answer Solved by Am
Contact: 0305-471661
Email: maniamaan2@

1 Edit Annotate Fill & Sign Convert All **2**



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EFFORT BY AMAAN

VU Med

CHE201:Grand Quiz

Question # 27 of 30 (Start time: 01:30:38 AM, 26 December 2020)

If ΔG is a _____ number the said reaction is spontaneous.

Select the correct option

☒	negative
☐	positive
☐	zero
☐	constant

Answer Solved by Amaan Khan
Contact: 0305-4716616
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CHE201:Grand Quiz

Question # 28 of 30 (Start time: 01:31:19 AM, 26 December 2020)

Diffusion is an example of

1 Edit Annotate Fill & Sign Convert All **5**



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Date: / /

Che 201, Timing 12:00 @ Nomi Rajput

MCQs

- * Thermodynamics studies the relationship between the _____ of system (macroscopic properties).
- * Two system in thermal equilibrium with each other have the same _____. (Temperature).
- * Calorimetry is _____. (Measurement Enthalpy)
- * _____ are liquid or solid with gas dispersed into them. (Foams).
- * _____ often used as a way or purify component. (Precipitations).
- * The rate at which reactant first brought to gether is called _____. (initial rate).
- * Boiling point of Ethanol _____. (78.5°C)
- * The number of particles involved in an elementary step is called _____. (molecularity).
- * The fan in a 'frost free' home freezer moves air around inside the freezer to _____. (Sublimate).

⇒ Short and Long Question

- ⇒ Name factor affect the rate of reaction?
- ⇒ How stopped flow effect mechanism?
- ⇒ What is Azeotropes?
- ⇒ What is Limitation of thermodynamics?



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MID/FINAL TERM 20__ (Year) Date: 15/2/24 Superintendent Signature: _____
Campus Code: _____ 4:30 Student Signature: _____
Course Code: Ahe 201 Student ID: _____

Calorimetry is _____
Irreversible process is one in which heat is _____
Gibbs free energy is _____
e.g. of dipole moment is _____
Interaction b/w solute particles with solvent is _____
Chemical kinetic is measurement of _____
Clock reaction method is used to determine rate of reaction

1st order reaction that depends on _____
_____ min. 3 molecules are necessary for reaction.

Short:
stop flow mechanism
elementary steps.
different types of system
precipitation
character monitored by rate reaction
eg:
effect of phase surface area & solvent effect
measurement of thermodynamic
viscosity by viscometer
fractional distillation
zone activation

Note: Submit this sheet to Superintendent, before leaving the Examination Center.

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from the top of the fractionating column cannot be pure ethanol.

- The best you can produce by simple fractional distillation is 95.6% ethanol.

- Chemical kinetics is the measurement of how quickly reactions occur.

continuously measuring.

- Continuous flow assays require a large amount of reagent, since it needs to flow continuously.

- It is very useful for measuring inhibition

Third Order Reaction

A reaction is said to be of third order if the rate is determined by the variation of three concentration terms. In other words, the

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to completely roll over the hill of a certain height.

- The faster the object moves, the more kinetic energy it has.

- In particular, a slow step in a mechanism determines the rate of a reaction.

Summary

- The number of particles involved in an elementary step is called the molecularity, and in general, we consider only the molecularity of 1, 2, and 3.

- As a chemical change takes place, the quantities of reactants and products change in a way that leads to a more negative free

- For simplicity, let us consider the reactions:

$$A + \text{other reactants} \xrightarrow{k} \text{products}$$

Where A is one of the reactants, and k is the rate constant.

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- This is the method that Manfred Eigen (Germany) pioneered when, in the early 1960's, he measured the rate constant of what was then the fastest reaction ever observed:



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- 150) The length of half-life will be constant, independent of initial concentration in
- 1st order reaction
 - Zero order reaction
 - Both
 - None
- 151) For first order reactions, the half-life depends
- on the reaction rate constant
 - concentration
 - rate of reaction
 - none
- 152) The length of half-life increase while the concentration of substrate constantly decreases.
- 1st order reaction
 - 2nd order reaction
 - 3rd order reaction
 - zero order reaction
- 153) For second order reactions, the half-life only depends on the
- initial concentration
 - the rate constant
 - both
 - none
- 154) Arrhenius performed experiments that correlated chemical reaction rate constants with---
- Temperature
 - Concentration
 - Reaction rate
 - None
- 155) As frequency factor A is related to molecular collision, it is--- dependent.
- Concentration
 - Temperature
 - Rate
 - None
- 156) What is the unit of E_a in Arrhenius equation.
- kJ/mol
 - kJ
 - kJ/mol^2
 - kJ/mol^3
- 157) Enzymes are proteins or ---molecules that provide alternate reaction pathways with lower activation energies than the original pathways
- RNA
 - DNA
 - Both
 - None
- 158) Enzymes ---the rate constant value.
- Decreases
 - Increases
 - None
 - Increases
- 159) Which of the following is the Eyring equation?
- $\ln K^\ddagger = -\frac{AS^\ddagger}{RT} + \frac{\Delta S^\ddagger}{R}$
 - $\ln K^\ddagger = -\frac{\Delta H^\ddagger}{RT} + \frac{R}{T}$
 - $\ln K^\ddagger = -\frac{\Delta H^\ddagger}{RT} + \frac{\Delta S^\ddagger}{R}$

- 160) The higher the activation enthalpy, the ---energy is required for the products to form
- More
 - Less
 - Very less
 - None
- 161) In general, a reaction proceeds ---if E_a and $\Delta H^\ddagger$ are small.
- Faster
 - Slower
 - Very fast
 - Very slow

162) E_a for a radical reaction is

- 0
- 1
- Negative
- Positive

7:23

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CHE201 Final...



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CHE 201 Current Final terms Fall 2020

M.C.Q

1. Thermodynamics studies the relationships between the _____ of a system. **Macroscopic properties**
2. Two systems in thermal equilibrium with each other have the same _____. **Temperature**
3. For a system in mechanical equilibrium, the pressure throughout the system is _____. **Uniform**
4. According to Charles law the volume of a fixed amount of gas, at constant pressure, is proportional to the _____. **Absolute temperature**
5. Calorimetry is _____. **Measurement of Enthalpy**
6. A measurement of the relative amounts of products and reactants present in a reaction at a given time is called _____. **Reaction quotient**
7. Number of moles of solute per liter of the solution _____. **Molar**
8. _____ are liquids or solids with a gas dispersed into them. **Emulsions**
9. The interactions between the solute particles and the solvent molecules is called _____. **Solvation**
10. _____ is often used as a way to purify a compound. **Precipitation**
11. In fractional distillation, the packing column is packed with glass beads to give _____. **Maximum possible surface area for vapor to condense on**
12. Continuous flow is a type of assay used for determining initial value and _____. **Inhibition value**
13. The rate at which the reagents are first brought together is called _____. **Initial rate**
14. The relaxation methods can be used to determine the _____. **Rate of the reaction**
15. A first-order reaction is a reaction that proceeds at a rate that depends on _____. **One reactant concentration**
16. Boiling point of Ethanol is _____. **78.5°C**

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17. The number of particles involved in an elementary step is called _____. **Molecularity**
18. The fan in a 'frost free' home freezer moves air around inside the freezer to _____. **Sublimate**
19. Extensive properties depend on the _____ of the system. **Parts**
20. A non-spontaneous is also known as _____. **Endergonic reaction**
21. _____ influences the physical properties of liquids. **Intermolecular forces**

Subjective

Qno1: What is the effect of catalyst on the activation energy? **2 marks**

If a catalyst is added to the reaction, the activation energy is lowered because a lower energy transition state is



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Petroleum refineries use distillation to separate crude oil

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2. Two systems in thermal equilibrium with each other have the same temperature.

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3. Two systems that are each found to be in thermal equilibrium with a third system will be found to be in thermal equilibrium with each other is called Zeroth law of thermodynamics.
The quantitative study and measurement of heat and enthalpy changes is known as thermochemistry.
The measurement of q or ΔH is generally known as Calorimetry.
6. Example of an irreversible process is Heat transfer.
7. The oxidation of gold is Nonspontaneous Processes.
8. The reaction quotient Q is a measure of the relative amounts of products and reactants present in a reaction at a given time.
9. Gibbs Helmholtz demonstrate that the free energy, not heat production, was the driver of spontaneous change in isothermal chemical reactions.
The molarity (M) of a solution is used to represent the amount of moles of solute per liter of the solution.
10. Foams are liquids or solids with a gas dispersed into them.
11. The interactions between the solute particles and the solvent molecules is called solvation.
12. Dissolution, the solute separates into ions or molecules, and each ion or molecule is surrounded by molecules of solvent.
13. Chemical kinetics is the measurement of how quickly reactions occur.
14. The fractionating column is packed with glass beads (or something similar) to give the maximum possible surface area for vapor to condense on.
15. The continuous flow assay offers a high degree of accuracy by allowing what amounts to a large number of readings at each time point.
16. The initial rate of reaction is the rate at which the reagents are first brought together.
17. Relaxation method: Many chemical reactions are complete in less than a few seconds, which makes the rate of reaction difficult to determine. In these cases, the relaxation methods can be used to determine the rate of the reaction.
18. Although a balanced chemical equation for a reaction describes the quantitative relationships between the amounts of reactants present and the amounts of products that can be formed.
19. A first-order reaction is a reaction that proceeds at a rate that depends linearly on only one reactant concentration.
20. In Third Order Reaction, the minimum number of molecules necessary for the reaction to take place is three.
21. Third order reaction does not depend on Product concentration.
22. The activation energy is the threshold energy that the reactant(s) must acquire before reaching the transition state.
23. The kinetic energy of a molecule is proportional to the velocity of the molecule.
24. Once started, chain reactions continue until the reactants are exhausted.
25. A mixture of hydrogen and chlorine will not react at all in the dark.
26. The concentrations of the different reactants will generally fall at different rates, depending on the stoichiometry.

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28. Chain reactions usually consist of many repeating elementary steps, each of which has a chain carrier.

2 / 15

29. Whether you use the differential rate laws or the integrated laws, you have to evaluate the order first.

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